WORKED SOLUTION

STEP III 2012, Question 11

Mechanics • Acceleration

STEP III 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q11

The key idea

Everything follows from the geometry of the loop. Once we know how long the stationary side and the moving side are when the particle has fallen x, we can write down the potential energy lost (using the centre of mass of each straight section) and the kinetic energy gained (only the particle and the moving side have speed). Energy conservation then gives v^2 as a function of x, and the acceleration comes from differentiating $\frac{1}{2}v^2$ with respect to x, since $v\,\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}v}{\mathrm{d}t}$.

Solution

The model and the geometry

We treat the rope as a uniform, inextensible, perfectly flexible line of mass per unit length

$$\rho = \frac{2M}{2L} = \frac{M}{L},$$

the particle as a point mass, and we take the bend at the bottom of the loop to have negligible size. Measure depths downwards from P.

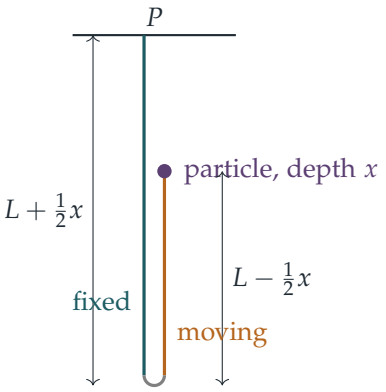
When the particle has fallen a distance x, the rope consists of two vertical sections joined at the bottom of the loop, at depth d say:

- the *fixed* section, from P down to the bottom, which is at rest and has length d;
- the *moving* section, from the particle (depth x) down to the bottom, which has length $d - x$.

The total length is $2L$, so $d + (d - x) = 2L$, giving

$$d = L + \tfrac{1}{2}x, \qquad \text{moving length } = L - \tfrac{1}{2}x.$$

(At the start, $x = 0$ and both sections have length L.)



Potential energy lost

Each straight section is uniform, so its centre of mass is at its midpoint. With depth h, a mass μ has potential energy $-\mu g h$. Write $D(x)$ for the sum of mass $\times$ depth over the whole system:

$$\begin{aligned} D(x) &= mx + \rho d \cdot \frac{d}{2} + \rho(d-x)\cdot\frac{x+d}{2} \\ &= mx + \frac{\rho}{2}\left(2d^2 - x^2\right) \\ &= mx + \rho\left(L^2 + Lx + \tfrac{1}{4}x^2 - \tfrac{1}{2}x^2\right) = mx + \rho\left(L^2 + Lx - \tfrac{1}{4}x^2\right). \end{aligned}$$

At the start $D(0) = \rho L^2$, so the potential energy lost is

$$g\big(D(x) - D(0)\big) = gx\left(m + \rho L - \tfrac{1}{4}\rho x\right) = \frac{gx}{L}\left(mL + ML - \tfrac{1}{4}Mx\right),$$

using $\rho L = M$.

Kinetic energy gained

The fixed section is at rest. The particle and every point of the moving section have speed v (as we are told to assume), so the moving mass is $m + \rho\left(L - \frac{1}{2}x\right)$ and

$$\text{KE} = \tfrac{1}{2}\left(m + M - \frac{Mx}{2L}\right)v^2 = \frac{1}{2L}\left(mL + ML - \tfrac{1}{2}Mx\right)v^2.$$

The system starts from rest, so energy conservation gives KE = PE lost:

$$\frac{1}{2L}\left(mL + ML - \tfrac{1}{2}Mx\right)v^2 = \frac{gx}{L}\left(mL + ML - \tfrac{1}{4}Mx\right),$$

that is,

$$v^2 = \frac{2gx\left(mL + ML - \frac{1}{4}Mx\right)}{mL + ML - \frac{1}{2}Mx},$$

as required. The denominator is never zero for $x < 2L$, since then $ML - \frac{1}{2}Mx > 0$, so $mL + ML - \frac{1}{2}Mx > mL \geq 0$.

The acceleration

For brevity write

$$A = mL + ML - \tfrac{1}{4}Mx, \qquad B = mL + ML - \tfrac{1}{2}Mx,$$

so that $\frac{1}{2}v^2 = gxA/B$, with $A' = -\frac{1}{4}M$ and $B' = -\frac{1}{2}M$ (differentiating with respect to x).

Ask yourself How do we get an acceleration from v^2 as a function of x?

By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}v^2\right) = v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}v}{\mathrm{d}t}$, which is the acceleration. This is valid while the particle is moving down, so x increases with t.

By the product and quotient rules,

$$\begin{aligned}\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{gxA}{B}\right) &= g\left[\frac{A}{B} + x\,\frac{A'B - AB'}{B^2}\right] \\ &= g\left[\frac{A}{B} + x\,\frac{-\frac{1}{4}MB + \frac{1}{2}MA}{B^2}\right] \\ &= g\left[\frac{A - \frac{1}{4}Mx}{B} + \frac{MxA}{2B^2}\right].\end{aligned}$$

The key simplification is that $A - \frac{1}{4}Mx = mL + ML - \frac{1}{2}Mx = B$, so the first fraction is exactly 1. Hence

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g + \frac{Mgx\left(mL + ML - \frac{1}{4}Mx\right)}{2\left(mL + ML - \frac{1}{2}Mx\right)^2},$$

as required.

Acceleration exceeds g

After release, $0 < x < 2L$. Then

- $B = mL + M\left(L - \frac{1}{2}x\right) > 0$, as shown above, so $B^2 > 0$;
- $A = B + \frac{1}{4}Mx > B > 0$;
- $M > 0$, $g > 0$ and $x > 0$.

So the extra term $\dfrac{MgxA}{2B^2}$ is strictly positive, and the acceleration is strictly greater than g for every x in $0 < x < 2L$. (At the instant of release, $x = 0$ and the acceleration is exactly g.)

Checks

- If $M = 0$ (a light rope) both formulae reduce to free fall: $v^2 = 2gx$ and acceleration g.

- Take $m = 0.3$, $M = L = 1$ and $x = 1$, with $g = 9.8$: then $A = 1.05$, $B = 0.8$, so $v^2 = 25.725$ and the acceleration is $9.8 + 9.8 \times 1.05/1.28 \approx 17.84\,\mathrm{m\,s^{-2}}$, well above g. Differentiating $\frac{1}{2}v^2$ numerically at this point gives the same value, and stepping the equation of motion forward from rest reproduces the v^2 formula.
- Units: mL, ML and Mx all have the units of mass $\times$ length, so the fractions are dimensionless and v^2 has units of gx.

Common slips

- Getting the moving length wrong: the bottom of the loop moves down by $\frac{1}{2}x$, not x, because the rope taken up by the falling side is shared between both sides.
- Giving the fixed section kinetic energy, or forgetting that it also changes its potential energy (it gets longer, so its centre of mass drops).
- Differentiating v^2 instead of $\frac{1}{2}v^2$, which gives twice the acceleration.

Beyond STEP

This is the classic "falling folded chain" problem, a system whose mass in motion changes with time. The energy-conserving model used here predicts that the end accelerates faster than g, and if $m = 0$ the speed grows without bound as $x \to 2L$, which is the idea behind the crack of a whip. Whether energy really is conserved at the bend has been debated in the research literature, and experiments with real chains show accelerations greater than g, as here. Variable-mass systems and the careful use of energy versus momentum are treated in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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